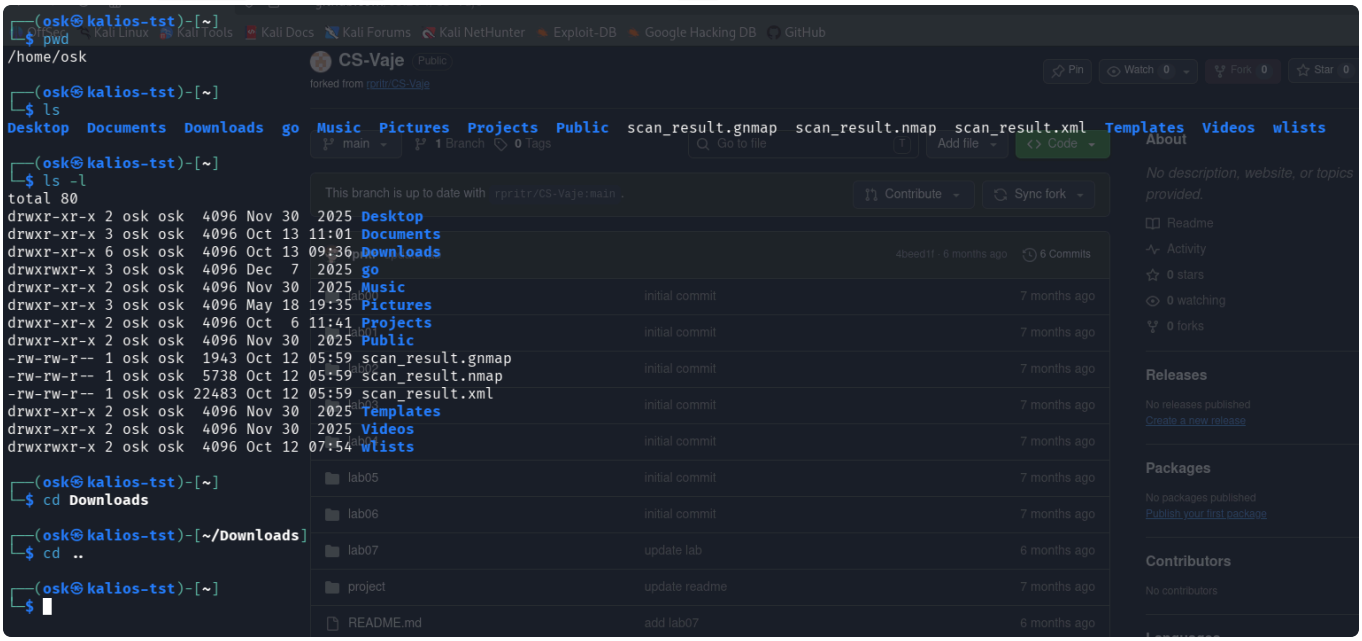
Lab00

#cyber-security

1

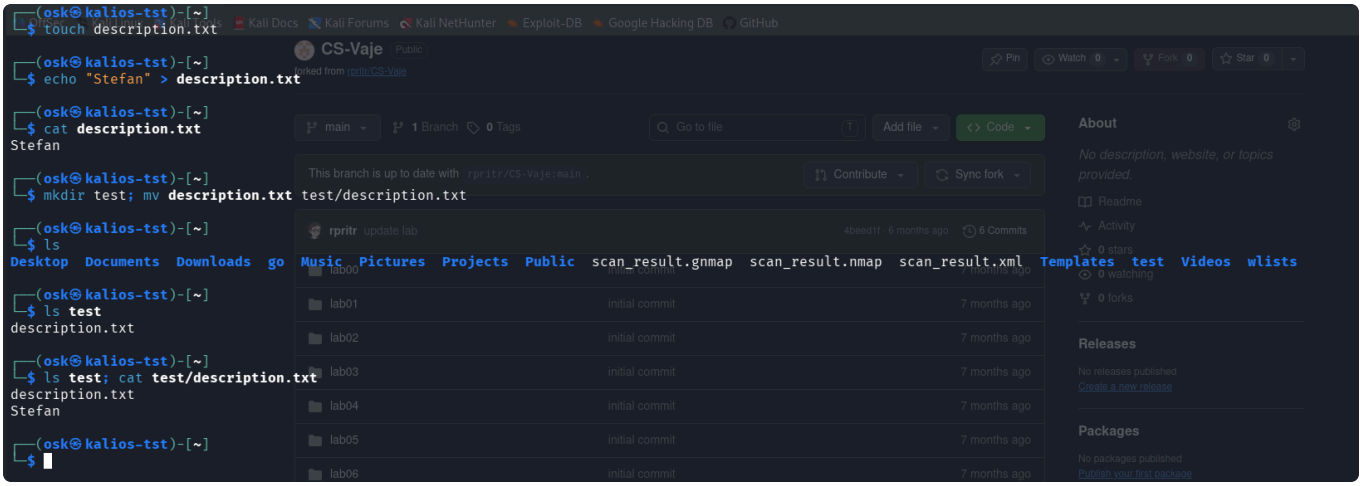
In this task, we first print out the current directory we are situated in via the command `pwd`.
Next, we display the contents of the current directory via `ls` and do the same but requesting more detail by adding an extra argument via `ls -l`.
Afterwards we move to another directory, let's take `Downloads` for example. We do this by writing the following command `cd Downloads`.
Finally, we can also move a single directory above in the filesystem hierarchy with the command `cd ..` where `.` means current and `..` means parent directory.



2

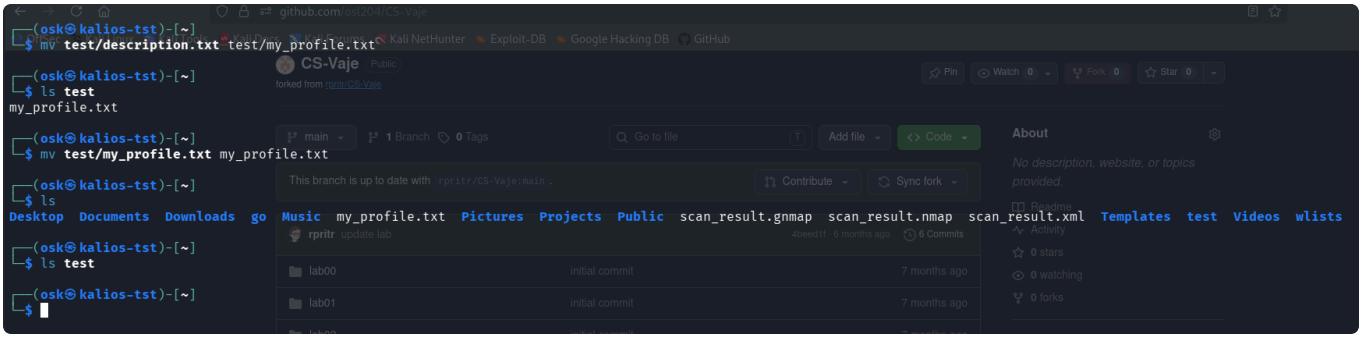
We first create a new empty file called 'description.txt' by running `touch description.txt`. We then write our name inside the file via the command `echo "Stefan" > description.txt`.
We can check if this worked by displaying the contents of the newly created description.txt file by running the command `cat description.txt`.
Next, we want to move this file to a new directory called 'test', first however we need to create this directory. We can do both by running `mkdir test; mv description.txt test/description.txt` where we separate the two commands with a `;` and having them execute sequentially.
We confirm that the folder has been created via `ls` in the current directory, we confirm that the file exists within the test directory via `ls test` then we confirm that the file exists unchanged

via `ls test; cat test/description.txt`.



3

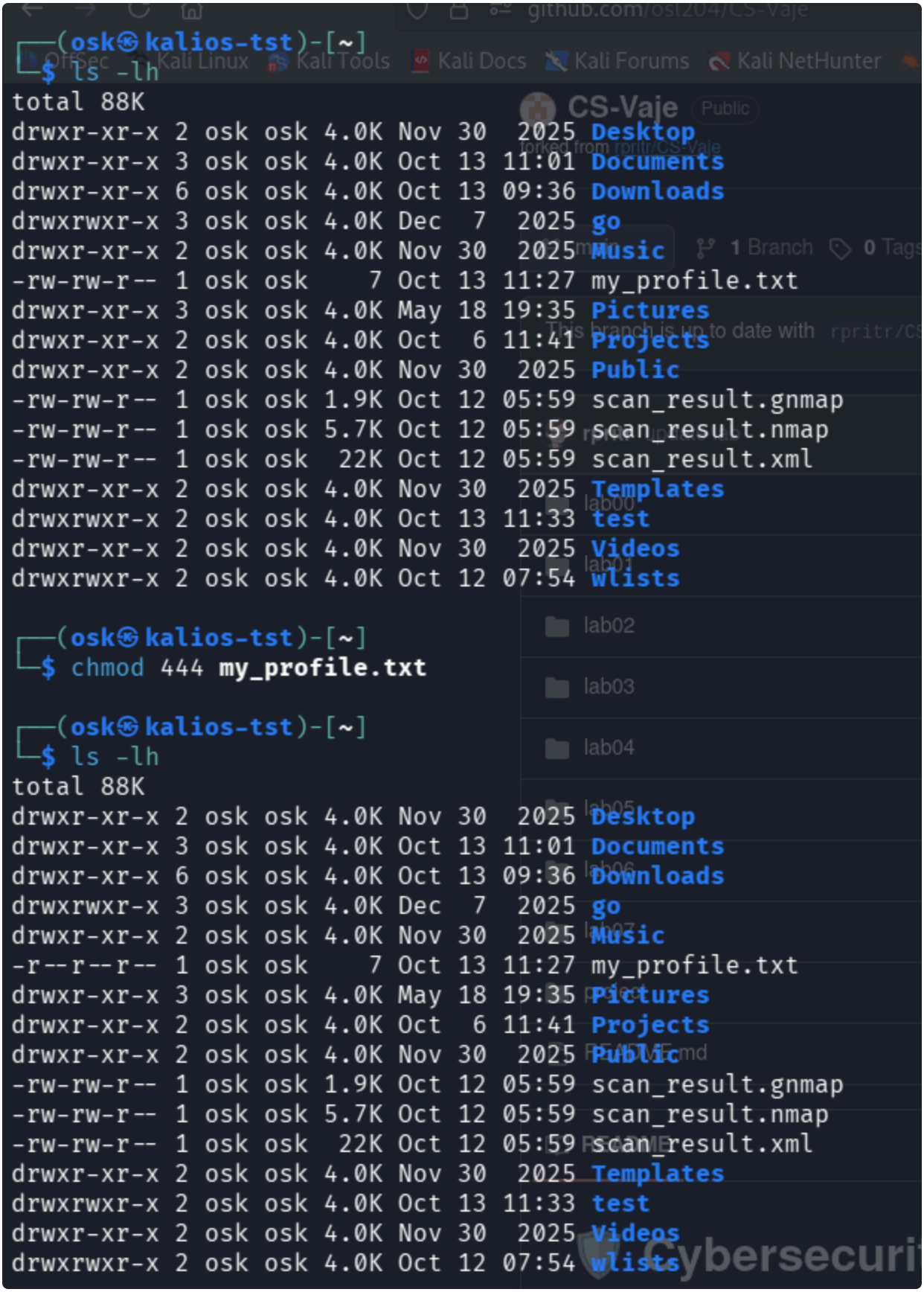
We rename the description.txt file to 'my_profile.txt' with the following command `mv test/descripton.txt test/my_profile.txt`. We can check that the renaming worked by writing `ls test`.
Next we want to move the renamed file to our home directory, we can do that by `mv test/my_profile.txt my_profile.txt` - Note: by writing another name for the file in the latter part of the `mv` command we can copy the file while also renaming it.
We then run `ls` to check if the move worked and `ls test` to see that that directory is now empty.



4

We can check all the files as before using `ls`, but we can also request additional information - such as the sizes of the files within the current as well as sub-directories by attaching `h`. So, we run the command `ls -lh` and see the following.
Next, we want to change the permissions of the 'my_profiles.txt' file so that everybody (owner-group-users) have only read privileges. To accomplish that, we run the following command `chmod 444 my_profile.txt`.

We can then verify this change by re-executing the previous command.



```
┌──(osk㉿kalios-tst)-[~]
└─$ ls -lh
total 88K
drwxr-xr-x 2 osk osk 4.0K Nov 30  2025 Desktop
drwxr-xr-x 3 osk osk 4.0K Oct 13 11:01 Documents
drwxr-xr-x 6 osk osk 4.0K Oct 13 09:36 Downloads
drwxrwxr-x 3 osk osk 4.0K Dec  7  2025 go
drwxr-xr-x 2 osk osk 4.0K Nov 30  2025 Music
-rw-rw-r-- 1 osk osk    7 Oct 13 11:27 my_profile.txt
drwxr-xr-x 3 osk osk 4.0K May 18 19:35 Pictures
drwxr-xr-x 2 osk osk 4.0K Oct  6 11:41 Projects
drwxr-xr-x 2 osk osk 4.0K Nov 30  2025 Public
-rw-rw-r-- 1 osk osk 1.9K Oct 12 05:59 scan_result.gnmap
-rw-rw-r-- 1 osk osk 5.7K Oct 12 05:59 scan_result.nmap
-rw-rw-r-- 1 osk osk  22K Oct 12 05:59 scan_result.xml
drwxr-xr-x 2 osk osk 4.0K Nov 30  2025 Templates
drwxrwxr-x 2 osk osk 4.0K Oct 13 11:33 test
drwxr-xr-x 2 osk osk 4.0K Nov 30  2025 Videos
drwxrwxr-x 2 osk osk 4.0K Oct 12 07:54 wlists

┌──(osk㉿kalios-tst)-[~]
└─$ chmod 444 my_profile.txt

┌──(osk㉿kalios-tst)-[~]
└─$ ls -lh
total 88K
drwxr-xr-x 2 osk osk 4.0K Nov 30  2025 Desktop
drwxr-xr-x 3 osk osk 4.0K Oct 13 11:01 Documents
drwxr-xr-x 6 osk osk 4.0K Oct 13 09:36 Downloads
drwxrwxr-x 3 osk osk 4.0K Dec  7  2025 go
drwxr-xr-x 2 osk osk 4.0K Nov 30  2025 Music
-r--r--r-- 1 osk osk    7 Oct 13 11:27 my_profile.txt
drwxr-xr-x 3 osk osk 4.0K May 18 19:35 Pictures
drwxr-xr-x 2 osk osk 4.0K Oct  6 11:41 Projects
drwxr-xr-x 2 osk osk 4.0K Nov 30  2025 Public
-rw-rw-r-- 1 osk osk 1.9K Oct 12 05:59 scan_result.gnmap
-rw-rw-r-- 1 osk osk 5.7K Oct 12 05:59 scan_result.nmap
-rw-rw-r-- 1 osk osk  22K Oct 12 05:59 scan_result.xml
drwxr-xr-x 2 osk osk 4.0K Nov 30  2025 Templates
drwxrwxr-x 2 osk osk 4.0K Oct 13 11:33 test
drwxr-xr-x 2 osk osk 4.0K Nov 30  2025 Videos
drwxrwxr-x 2 osk osk 4.0K Oct 12 07:54 wlists
```

5

We can check the user we are currently active as via the command `whoami`.
Next, we can check the current directory's size via the command `du -hs` (providing an argument such as the full path or even `./` | `.`) will have the same result.).
We can also check the total size as well as available space via `df -h`

